SAMUEL MEKONNEN ERKUYE

Software Engineer • AI Automation Specialist • Self-Taught Full-Stack Developer

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SUMMARY

Self-taught full-stack engineer who builds production systems end to end and automates the manual work around them. In 2026 I shipped 10 projects with 550+ commits: a multi-tenant shuttle-management platform with Flutter apps and a routing engine, a wallet platform whose deposits are verified automatically from SMS, real-time multiplayer apps on Socket.io, and a browser extension that cut a 15-click call-center workflow to one click. I work AI-native: I write the spec, direct AI coding agents (Claude Code), review every change and back it with tests (175+ test files across my repositories). Two-plus years in B2B sales and operations mean I build what the business actually needs, and I can explain it to the people who use it.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL, Dart, Kotlin, HTML/CSS
- **Frontend & mobile:** React 19, Next.js 16 (App Router), Vite, Tailwind CSS, PWAs, Telegram Mini Apps, Flutter (Android/iOS), native Android (Kotlin)
- **Backend & data:** Node.js, Express 5, Socket.io, REST/OpenAPI, Prisma, PostgreSQL/PostGIS, SQLite, Redis, BullMQ, AWS S3
- **AI & automation:** Claude Code and AI coding agents, ChatGPT, prompt engineering, browser extensions (Manifest V3), Telegram bots, job schedulers and queues, SMS/webhook pipelines, Excel/CSV data pipelines
- **Security:** Argon2id, JWT with rotating refresh tokens, HMAC-signed webhooks, RBAC, row-level security, OTP login, rate limiting
- **Delivery:** Git/GitHub, Vitest, ESLint, Docker Compose, Vercel, Railway, Render, Neon, Cloudflare Tunnels

SELECTED PROJECTS

Derash: Employee Shuttle Management Platform | built for an MMCY Tech pilot — **Oct 2026**

TypeScript • Express 5 • PostgreSQL/PostGIS • Redis/BullMQ • Socket.io • React PWA • Flutter

- Built a multi-tenant platform that gets employees home after shifts: dispatcher dashboard, rider and driver apps (Flutter, Android/iOS) and a provider console, in English and Amharic with Ethiopian time.
- Wrote a deterministic assignment engine: k-means zone clustering, smallest-vehicle fit, trip consolidation and nearest-neighbour + 2-opt route ordering, with incremental re-planning for absent drivers and breakdowns.
- Automated the whole shift lifecycle (T-60 / T-15 / T-5 autopilot) and a push → Telegram → SMS notification cascade with delivery receipts; live GPS tracking over Socket.io.
- Production API with row-level-security tenancy, rotating refresh tokens, audit log, idempotency keys and rate limits; 49 test files including an end-to-end shift-end flow.

Fyel: Digital Wallet & Gaming Platform | in production, 2-person team — **Aug – Oct 2026**

Next.js 16 • React 19 • PostgreSQL • Prisma 7 • Kotlin (Android) • Telegram Bot API

- Built the money core: every balance change recorded with before/after values, exact decimals, and per-wallet row locks that stop concurrent double-spends (covered by concurrency tests).
- Automated deposit verification: a Kotlin Android app forwards the agent handset's payment SMS, and the server credits deposits automatically when transaction ID, account and amount match. Only mismatches reach a human.
- Integrated 3,000+ third-party titles through a provider API with HMAC-SHA256-verified, idempotent callbacks; Telegram Mini App sign-in and bot notifications; admin dashboard with audit trail; 50 test files.

Dialer Assistant: Call-Center Browser Automation | used daily at MMCY Tech — **Oct – Nov 2026**

JavaScript • Edge/Chrome Manifest V3 extension • SheetJS • Zoiper softphone integration

- Turned a manual shareholder-call workflow into one click: imports an Excel phone → control-number file, finds the caller instantly, and runs a three-pass disposition workflow on its own (~15 manual clicks per call before).
- Receives the caller's number straight from the Zoiper softphone; automates scheduled breaks and clock in/out, and keeps working with the screen locked using background alarms.

JAH: High-Concurrency Gaming & Fintech Platform — **Oct 2026**

TypeScript monorepo (pnpm, Turborepo) • Express • Prisma • PostgreSQL • Redis • Socket.io

- Player site, back office, agent and partner portals plus a worker service. Built an append-only double-entry ledger with integer (BigInt) arithmetic and rotating JWT refresh-token families with theft detection; 43 test files.

Tebiban School: Ethiopian Music E-Learning Platform **Jun – Nov 2026**

Next.js 16 • TypeScript • Prisma 7 • PostgreSQL (Neon) • Resend • Vercel

- Migrated the platform from Supabase to a server-first Prisma architecture with self-hosted authentication (Argon2id, signed httpOnly sessions, Google sign-in). Fixed 7 security issues along the way, including a committed service-role key and a client-side admin-escalation path.

Real-Time Multiplayer Bingo (Telegram Mini App) | extended an open-source base **Sep – Oct 2026**

React • Node.js • Socket.io • SQLite • Redis

- Four live rooms with SHA-256 committed (provably fair) draws, Telegram WebApp HMAC login, agent and admin dashboards, plus security fixes for agent scoping and payout locking; 37 test files.

More **2026**

- DawitType: Amharic/English typing trainer for the 151 Psalms with Amharic keyboard mappings and synthesized sound (React, Vite). • Casino Suite: 10 canvas browser games with RTP controls (Node, WebSockets). • Startup Task Tracker (Next.js, Supabase auth). • SDR portfolio site (deployed on Vercel).

EXPERIENCE

Business & Sales Development Representative | MMCY Tech **Apr 2024 – Present**

- Built the Dialer Assistant to automate repetitive call-center work for my team, and proposed and prototyped the Derash shuttle platform for MMCY's employee transport.
- Run B2B outbound (150–250+ dials a day), demos and proposals for MMCY's software and IT services.

Sales Development Representative (Part-time) | 3BL Studios **Jun – Oct 2026**

- First sales hire. Built the sales function from scratch and automated its busywork: lead-list cleanup, CRM workflows, follow-up triggers and activity reporting.

IB Computer Science TA & Tech Support | SOS Hermann Gmeiner International College, Ghana **Aug 2022 – Jun 2023**

- Taught programming to IB DP/MYP students and ran the Programming & Robotics Club; maintained servers, LAN, laptops and AV systems, and supported users in person and remotely.

Earlier **2021 – 2024**

- Assistant Business Developer (EAFMG); Marketing Specialist (Empire Syndicate, U.S.); co-founder of two ventures; CFO & AV Production Manager of an African Leadership Academy student enterprise.

EDUCATION & LANGUAGES

Advanced Diploma | African Leadership Academy, Johannesburg **2022**

High School Diploma | Bahir Dar STEM Center, Ethiopia **2020**

Self-taught software engineering through building and shipping real products. • Languages: English (C1), Amharic (native)